

Multi-Agent Autonomous Forensic Swarm & Fed SR 11-7 Cryptographic Audit Pipeline

1. Investigator Agent

- GNN Subgraph Extractor
- Causal 2-hop ego-net
- Multi-hop cycle synthesis
- Extracts entity KYC traits



2. SAR Drafter Agent

- FinCEN Form 111 XML
- Narrative auto-drafter
- Smurfing breakdown
- Cross-entity flow audit



3. Compliance Auditor

- BSA/AML rule validator
- Sanction list checker
- Conformal certifier
- Auto-escalate Tier 2



4. Fed SR 11-7 Audit Logger

- SHA-256 Merkle chain
- Nonce & timestamp seal
- Tamper-proof logs
- Zero-Knowledge export